

NIKKO DUMRIQUE

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SUMMARY

Backend engineer specializing in cloud infrastructure, serverless systems, and production AI applications. Designs and maintains MongoDB and S3 data schemas (5TB) with automated maintenance and data-recovery scripts, and builds cost-efficient AWS platforms and RAG/LLM pipelines that make a 50TB+ analytics warehouse accessible for ML.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Cloud & Infrastructure: AWS (Lambda, Lambda@Edge, S3, CloudWatch, API Gateway, Cognito), GCP, Serverless Architecture

AI/ML: RAG Pipelines, AWS Bedrock, Qdrant (Vector Search), Embeddings (Cohere Embed V4), Retrieval Evaluation (P@1, MRR), Model Deployment (Flask, Gunicorn), LightGBM

Data Engineering: ETL Pipelines, Airflow, Redshift, MongoDB, S3 Data Lakes, Data Modeling

Tools & DevOps: Docker, Git, CI/CD (GitHub Actions, CircleCI), Linux

EXPERIENCE

Senior Software Developer (Data Infrastructure & Analytics)

Jan 2024 – Present

Responsive Ads

Vancouver, BC

- Architected and led development of a production RAG pipeline on AWS Bedrock and Qdrant, leveraging Cohere Embed V4 semantic retrieval across 400K+ ad creative embeddings to automate structured JSON layout generation; established retrieval benchmarks (P@1, MRR).
- Managing a team executing Lambda backend modernization with blue-green deployments, a structured error framework, and enhanced CloudWatch observability to improve deployment safety and system reliability.
- Built backend data infrastructure for a 55TB Redshift warehouse, 3TB S3 data lake, and 700GB MongoDB dataset, enabling scalable analytics and ML/AI training.
- Enabled a UBC Masters of Data Science capstone team by architecting isolated cloud environments (IAM, S3, MongoDB, GCP), building custom Python tooling for data access, and guiding data cleaning, validation, and normalization practices.
- Identified cost and performance inefficiencies from billing and Zendesk analytics, then implemented automated management scripts (S3 lifecycle policies, GCP Compute Engine cron jobs, Lambda schedulers) that reduced storage footprint by 42% and improved system stability.

Backend Software Developer (Data Infrastructure & Analytics)

Sept 2023 – Jan 2024

Responsive Ads

Vancouver, BC

- Led development of a platform-wide error tracking and auto-recovery system, reducing monthly engineering support tickets by 86% and improving system reliability and response time during critical failures.
- Analyzed Zendesk support data to design automated error recovery workflows with self-healing capabilities, preventing cascading failures during rapid product scaling and reducing MTTR by 74%.
- Architected the migration of analytics tracking from EC2 to Lambda@Edge, cutting monthly infrastructure costs by approximately \$7,500 (65% reduction) and improving response latency by 9%.
- Built an in-house event-driven image optimization service (API Gateway, Cognito, Lambda, S3) to replace an unreliable third-party API, reducing monthly costs from \$5K to \$100 while improving uptime.

Software Developer (Full Stack)

May 2020 – Aug 2021

Emerging Media Labs

Vancouver, BC

- Developed automated ETL pipeline supporting multiple data formats (XML, JSON, CSV) with data validation and quality checks, reducing manual processing time by 82%
- Conceptualized, coded, and documented a SQL database and accompanying Python scripts to handle large amounts of data for OLTP usage, resulting in a 33% improvement in first-byte latency.

PROJECTS

Data Engineer: Financial Data Pipeline

Airflow, LightGBM, Docker

- Constructed financial data pipelines using Apache Airflow, Flask, Docker, and Python to predict trading volume from historical data, improving forecasting RMSE by 20% with parallel thread-pool ingestion that cut execution time by 47%.
- Deployed a LightGBM regression model through a Flask API using Gunicorn and CircleCI, enabling real-time predictions and reducing deployment time by 52%.

EDUCATION

University of British Columbia

B.Sc in Physics, Minor in Data Science

Vancouver, BC

Sept 2017 – May 2022